FEUP Universidade do Porto
Faculdade de Engenharia

Enterprise Information Systems

TeamWork 2026/2027

Sprint 1

Team GO3

Team

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Task description

The development of Sprint 1 was divided among all team members according to the different frontend and system requirements. The team collaboratively discussed the system requirements, page structure and user interfaces before distributing the implementation tasks. Each member was responsible for specific components of the project, as described below.

Maria Torrão da Costa (up201806494)

Responsible for the design and prototyping phase of the Courier Management System interface. Developed the mockups for all required pages, including the Home Page, Login/Register Area, Client Page, Courier Page, Staff Panel, Admin Panel and Team Page. Defined the visual layout, navigation structure, user experience flow and overall design consistency of the website, providing the foundation for the frontend implementation. Additionally, was responsible for compiling, organizing and writing the Sprint 1 report, ensuring the documentation accurately reflected the work developed by all team members throughout the sprint.

Alexandre Gorra (up202207016)

Responsible for the implementation of the Home Page and Courier Page. Integrated the Leaflet.js interactive maps required by the specification, including marker visualization and coordinate handling. Implemented JavaScript form validation mechanisms and contributed to the organization of the project structure, ensuring maintainable and structured source code. Additionally, connected the different application pages, improved overall consistency between interfaces by standardizing styles across the website, implemented several final adjustments to meet project requirements, and contributed to the project setup by providing an automated script for local deployment and execution.

Aaron Sahni Mayal (up202603531)

Developed the Login/Register and Client pages, including the corresponding CSS styling and frontend functionality. Implemented JavaScript-based form validation and integrated OpenStreetMap/Leaflet features required by the project specifications. Also developed the order status table used for delivery tracking and information visualization.

Michał Mateusz Mętel (up202602038)

Responsible for the overall coordination of Sprint 1 activities as team leader. Managed the GitHub repository structure and version control process, ensuring the integration of team contributions throughout the sprint. Developed the Team Page containing the academic identification of all project members and created the Use Case Diagram to represent the main system functionalities and user interactions. Additionally, coordinated task allocation among team members and was responsible for preparing and submitting the final project deliverables for evaluation.

Hugo Miguel de Ângelo Gomes (up202208777)

Responsible for the design of the Entity Relationship (ER) Diagram and the implementation of the Admin Page. The ER diagram was created to support the future development of the database layer and to establish a structured representation of the application's data model.

Diogo Teixeira Pinto (up202205296)

Responsible for the implementation of the Staff Page(HTML, CSS and JavaScript). Developed the courier and vehicle approval tables, where the validity of insurance and inspection documents is calculated automatically from their expiry dates and displayed through colour-coded indicators. Implemented the review and reject actions for courier applications using simulated data.

Work limitations

As Sprint 1 focused primarily on frontend development, several functionalities remain in a prototype stage. Database integration and persistent data storage have not yet been implemented, meaning that application data is not retained after refreshing the page.

Although the mapping functionality has been integrated using OpenStreetMap and Leaflet, route visualization is currently represented by a simple line between two locations and does not yet include advanced routing features.

The Entity Relationship Diagram was developed during this sprint; however, its integration into the implementation phase will only occur in future iterations when backend development begins.

Additionally, the application currently operates with static data and simulated interactions. User authentication, order persistence, vehicle approval workflows and dashboard statistics are represented only at interface level and are not yet connected to a backend system. Some interface components were developed independently by different team members and may require further visual standardization during future iterations to ensure complete consistency across the platform.

Observations

During Sprint 1, the team focused on developing the frontend prototype while following the project requirements and predefined page structure. Since different pages were implemented by different team members, some visual and layout differences emerged between interfaces. Although efforts were made to maintain a consistent design language, additional refinement and standardization will be carried out in the next sprint to improve overall visual coherence. The team also discussed improvements related to the presentation of member information on the Team Page, particularly regarding the sizing and organization of profile photographs to ensure a more uniform and professional appearance. The GitHub proved essential for task coordination, version control and integration of the work developed by each team member.

Furthermore, in addition to the mandatory requirements, the team took the initiative to implement several extra features to improve the system's robustness, security, and developer experience:

- Advanced Geocoding and Location Validation: The Leaflet.js interactive maps were enhanced to map GPS coordinates directly to actual street addresses (and vice versa). Additionally, we implemented a validation mechanism that checks if the selected coordinates are valid, preventing users from placing markers in unreachable places (e.g.,the middle of the ocean) **(see Figure 1)**.
- Local Development Environment: To streamline our development and testing workflow, a custom Python script was created. This script automatically starts a local server and runs the frontend directly in the browser via localhost, ensuring all paths and modules load correctly without CORS issues.
- Security Prototyping (Captcha): A Captcha verification component was integrated into the Login/Register interface. While the verification logic is currently static due to the lack of a backend in Sprint 1, the UI is fully prepared for the functional security integration planned for Sprint 2 **(see Figure 2)**.

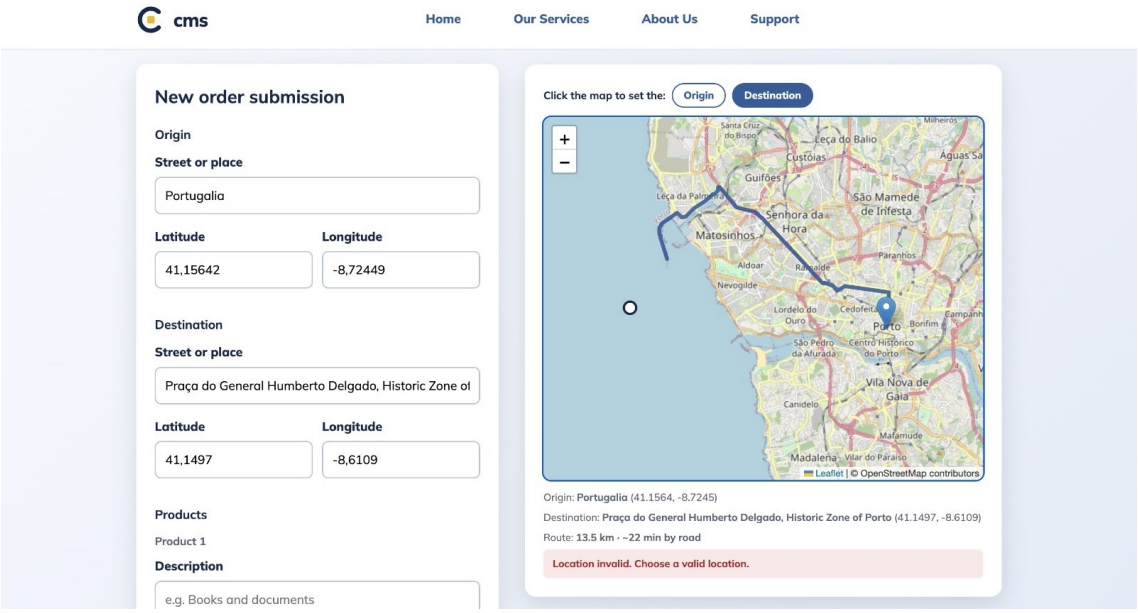


Figure 1: Location validation mechanism preventing the selection of unreachable coordinates.

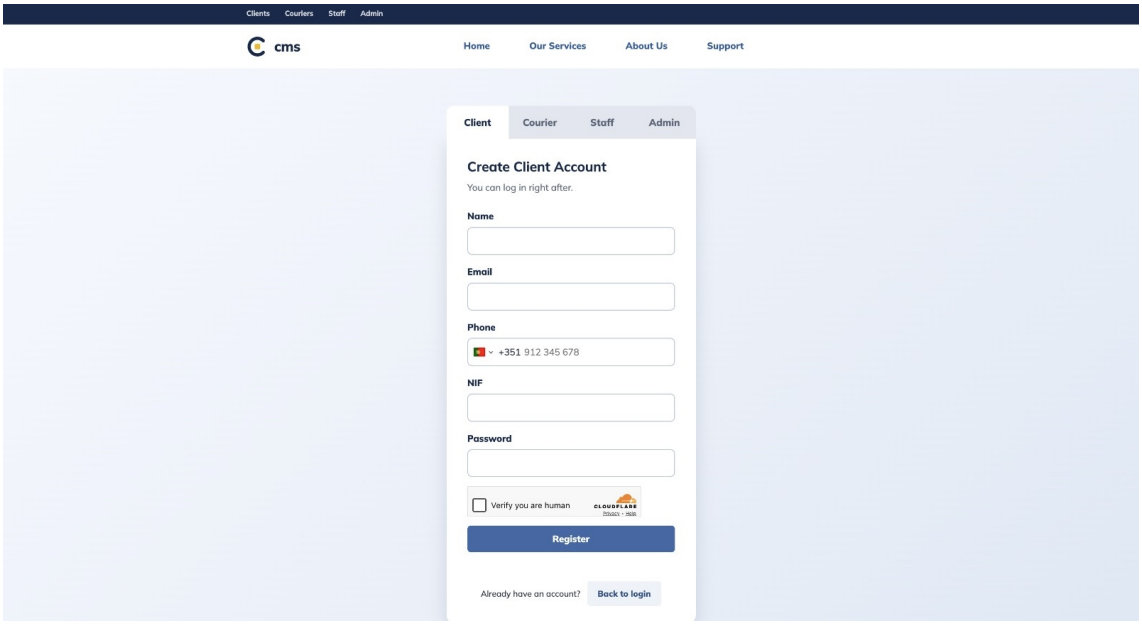


Figure 2: Static Captcha verification UI integrated into the Login/Register page, running on a local development server.